

Title: AI-Automated Classification of Hospital Safety Incidents

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Background and Motivation

The hospital's commitment to patient safety and continuous quality improvement necessitates a robust and up-to-date method for classifying safety incidents. Currently, the best practice for classifying hospital safety events is referencing the definitions and algorithms defined by the Healthcare Performance Improvement (HPI) methodology. However, manual HPI classifications of safety incidents are extremely time-consuming and require considerable expertise; hospital quality teams depend on trained professionals to review each safety incident report in detail, taking into account multiple contextual factors to classify the incident's severity. This project aims to harness machine learning and large language models combined with AI to automate the safety incident classification process.

Problem Statement

Manual classification of hospital safety incidents using the HPI methodology is highly time- and resource-demanding, limiting the ability of hospitals to quickly and accurately monitor patient safety events. We hope to design and develop a web-based tool that automates this process to assist hospitals in efficiently classifying safety incident reports, in addition allowing multilanguage inputs.

Data Source

Source: As safety event classification is a highly specialized task that requires expertise in medicine, we partnered with a Canadian hospital (name has been hidden for privacy concerns) and acquired access to ~100 annotated records of safety events following the guidelines published by PressGaney in 2022.

Description & Key Attributes: The raw dataset was presented in both structured and unstructured formats. The unstructured part includes a detailed textual description of the safety event while the structured part provides a manually curated pre-defined safety event classification level (e.g., SSE1: death) and a more nuanced taxonomy event type (e.g., CM8: delay in diagnosis or treatment).

Relevance: For the purpose of our project, both unstructured textual and structured classification labels are vital to fine-tuning a language model. Natural language processing techniques including tokenization and embedding can be applied to extract features from the unstructured textual data, which will serve as the foundation for fine-tuning a language model for a classification task. The structured data will be subsequently used as labels during the fine-tuning phase.

Data Quality Concerns: Despite the fact that the safety event records were annotated by medical professionals, challenges have to be acknowledged. The training set size might be most concerning given that we only have ~100 labeled cases as of now. The team has been in contact with the Hospital administration with the hope of acquiring more labeled cases. Furthermore, deviation is inevitable among the provided manually labeled cases, even by trained professionals. We have planned an integrated real-time feedback system which allows users with higher authorization levels to re-annotate and modify reported safety events with labels. The language model will iteratively and continuously learn from the updated annotation.

Scope and Objectives

Design and build a web-based tool that uses large language models to help hospitals and healthcare staff efficiently and accurately classify safety incident reports following the HPI methodology.

Objectives:

1. Collect and preprocess a diverse dataset of hospital safety incident reports.
2. Apply de-identification to remove Protected Health Information (i.e. patient name, DOB, medical record ID, etc.) and standardize input formats.
3. Develop LLMs to classify severity levels using the HPI SEC framework, and also fine-tuning the model for accuracy, aligning domain-specific terminology, and incorporating feedback from reviewers and user feedback for active learning.
4. Enable voice-to-text transcription and support multilingual input for diverse hospital environments.
5. Extract root causes of safety events (e.g. workflow gaps, communication failures, etc.). Generate structured reports summarizing classification and severity outcomes, and design dashboards for time trends, hotspot analysis, heatmaps, and SSER computation.

Minimum Components for a Good Project:

- Large Data: Collect hospital safety incident reports from publicly available datasets (e.g. incident reporting resources, AHRQ Patient Safety Network), and simulate synthetic cases for data augmentation.
- Scalability: The backend must support multiple hospital departments and user roles simultaneously (clinicians, administrators, safety officers), and manage increasing volumes of reports without performance degradation.
- Complex Models: LLM-based classification model and fine-tuning method.
- Computationally Expensive Inference: LLM-driven classification demands high-performance computing resources. We will train a lightweight model to mimic the outputs of a larger LLM and convert weights to lower precision to reduce GPU load.

Learning Emphasis: This project applies advanced LLM and fine-tune methods to incident classification, severity grading, and summarization, integrated with MLOps for deployment and iterative refinement, directly related to course concepts.

Application Mock Design:

- Input screen: voice/text upload, language selector.
- Processing screen: loading indicator.
- Output screen: classification label, rationale, recommended action.
- Dashboard view: trends, heatmaps, severity distribution.

Research and Development:

[1] Press Ganey. (n.d.). HPI SEC & SSER: Patient safety measurement system for healthcare. Press Ganey.

[2] Agency for Healthcare Research and Quality. (n.d.). AHRQ Patient Safety Network (PSNet). U.S. Department of Health & Human Services. Retrieved September 23, 2025, from <https://psnet.ahrq.gov>

[3] De Micco, F., Di Palma, G., Ferorelli, D., De Benedictis, A., Tomassini, L., Tambone, V., Cingolani, M., & Scendoni, R. (2025). Artificial intelligence in healthcare: Transforming patient safety with intelligent systems—A systematic review. *Frontiers in Medicine*, 11, 1522554. <https://doi.org/10.3389/fmed.2024.1522554>

Fun Factor: Working on real hospital safety challenges with AI makes the project meaningful and engaging. The interactive dashboard and feedback loop add a creative element, allowing the team to combine technical problem-solving with impactful healthcare applications.

Milestones:

Data collection and preprocessing – **Tuesday 09/30/2025**

LLM development for classification and severity grading – **Tuesday 10/14/2025**

Dashboard and SSER analytics implementation – **Sunday 10/26/2025**

Feedback and active learning integration – **Thursday 11/6/2025**

Final testing and deployment – **Thursday 11/20/2025**